

Churn is the revenue ceiling: from diagnosis to control

93,358 customers · 99,992 orders · 2016-09 → 2018-10 · XGBoost + causal uplift + Power BI

74.1%

customers churned

3.85%

satisfied → repeat step

R\$9.77M

revenue held by churned customers

R\$0.7–1.9M

annual GMV upside, elastic band

Revenue is capped by retention, not by traffic: the platform converts almost every order to delivery and review, then loses nearly every satisfied customer.

****What this deck proves**** — where the funnel breaks, why customers leave, what it costs, and which interventions pay back.

****Method**** — 7-node order funnel, RFM, cohort retention, gradient boosting with TreeSHAP attribution, causal identification (PSM + DoWhy DAG with refutation checks), NLP on reviews.

****Scope of the evidence**** — 99,992 orders and 93,358 customers from a Brazilian marketplace, 2016-09 → 2018-10, plus a 10-page Power BI monitoring dashboard.

****What is out of scope**** — acquisition-channel economics, pricing tests, seller negotiation.

Zhao Zhang · Data analyst · 2026-09 · figures reproducible from the project repository

zzhangzhao@hotmail.com · [Upwork profile](#) 

74% of customers churn after they are satisfied

Answer first · 4 supporting findings · 3 actions requested

Olist loses 74% of its customers and 73.6% of historical revenue at the satisfied-to-repeat step; late delivery is causally identified as a driver, so logistics is the controllable lever.

- 1 The journey breaks at the end.** 99% of orders convert to delivery and 96% are reviewed, but only 78.9% of reviews are positive and just 3.85% of those customers buy again.
- 2 Repeat customers are worth 2.0× more.** R\$277.92 vs R\$138.05 per head, so every lost repeat purchase compounds against the next quarter's revenue.
- 3 Logistics causes it, not merely correlates.** Matching comparable customers cuts the raw churn gap from 6.4pp to 3.9pp and the satisfaction gap from 1.72 to 0.46 points; all three refutation checks pass.
- 4 The fix is measurable, but the payback is a band.** Logistics pays back 0.95–4.34× — below cost at the fitted elasticity, 4.34× at the assumption; the 90-day control group settles which applies.

Actions requested

- Approve the logistics-first retention sprint for the next 90 days (S25).
- Adopt the scored watchlist as the operating list for CRM and service.
- Fund the A/B measurement so the next review uses measured uplift, not assumptions.

Three quarters of the revenue base must be re-earned each year; retention is the revenue plan, not a satisfaction project. Every number above rests on one explicit caliber — churn means no purchase for 180 days (S22).

97% of customers never buy twice: 32pp below benchmark

Repeat rate, all customers, snapshot 2016-09 → 2018-10

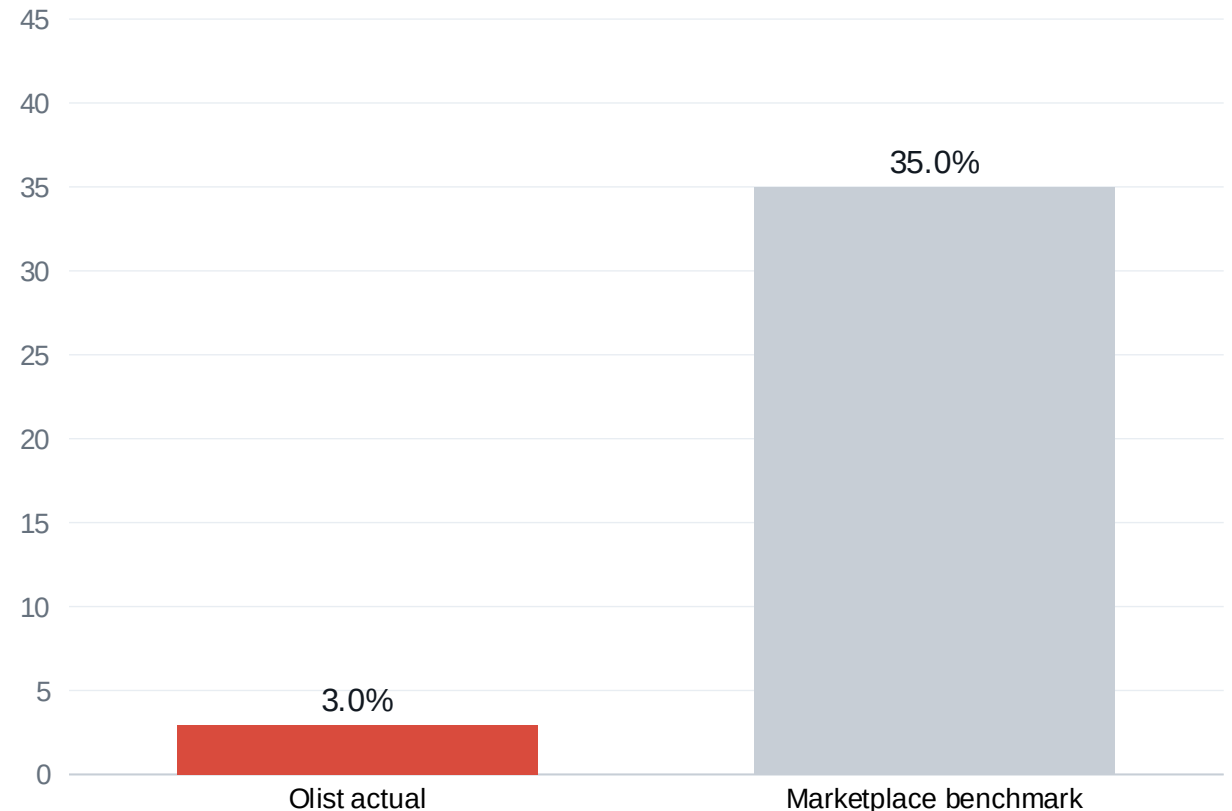
2.96%

Repeat rate across 93,358 customers

97% of customers ordered exactly once. Marketplace benchmarks sit at 30–40%, so the platform is not merely below average — it has no repeat habit to build on.

Counter-check: repeat buyers exist and behave normally — they spend 2.0× more (R\$277.92 vs R\$138.05). The problem is volume, not value. Caliber note: 803 of those 2,766 repeat buyers ordered only on a single day (split carts), so on true cross-period repurchase the rate is 2.10% — see S22.

Growth cannot compound until the first repurchase happens; that single step is worth more than any acquisition push.



Above benchmark on every step but repeat: 3.85%

17 comparisons: funnel nodes, retention windows, customer classes, annual churn

Funnel step vs industry benchmark

Node	Benchmark	Olist	Gap
Approved	99%	99.8%	+0.8pp
Shipped	95%	98.3%	+3.4pp
Delivered	95%	98.8%	+3.8pp
Reviewed	50%	99.3%	+49.3pp
Satisfied (≥4)	70%	78.9%	+8.9pp
Repurchased	30%	3.9%	-26.1pp

Retention windows vs healthy

Node	Benchmark	Olist	Gap
30 -day retention	60%	0.0%	-60.0pp
90 -day retention	40%	6.4%	-33.6pp
180 -day retention	30%	25.9%	-4.1pp
365 -day retention	25%	68.7%	+43.7pp

Annual churn vs other models

Node	Benchmark	Olist	Gap
vs e-commerce	25%	74.1%	+49.1pp
vs marketplace	40%	74.1%	+34.1pp
vs subscription	5%	74.1%	+69.1pp

-26.1pp

Repurchase vs the 30% norm

3.85% actual

-33.6pp

90-day retention vs 40%

6.4% actual

+180%

New buyers vs 30-40%

98.07% actual

-99.5%

Consecutive repeat vs 30-100%

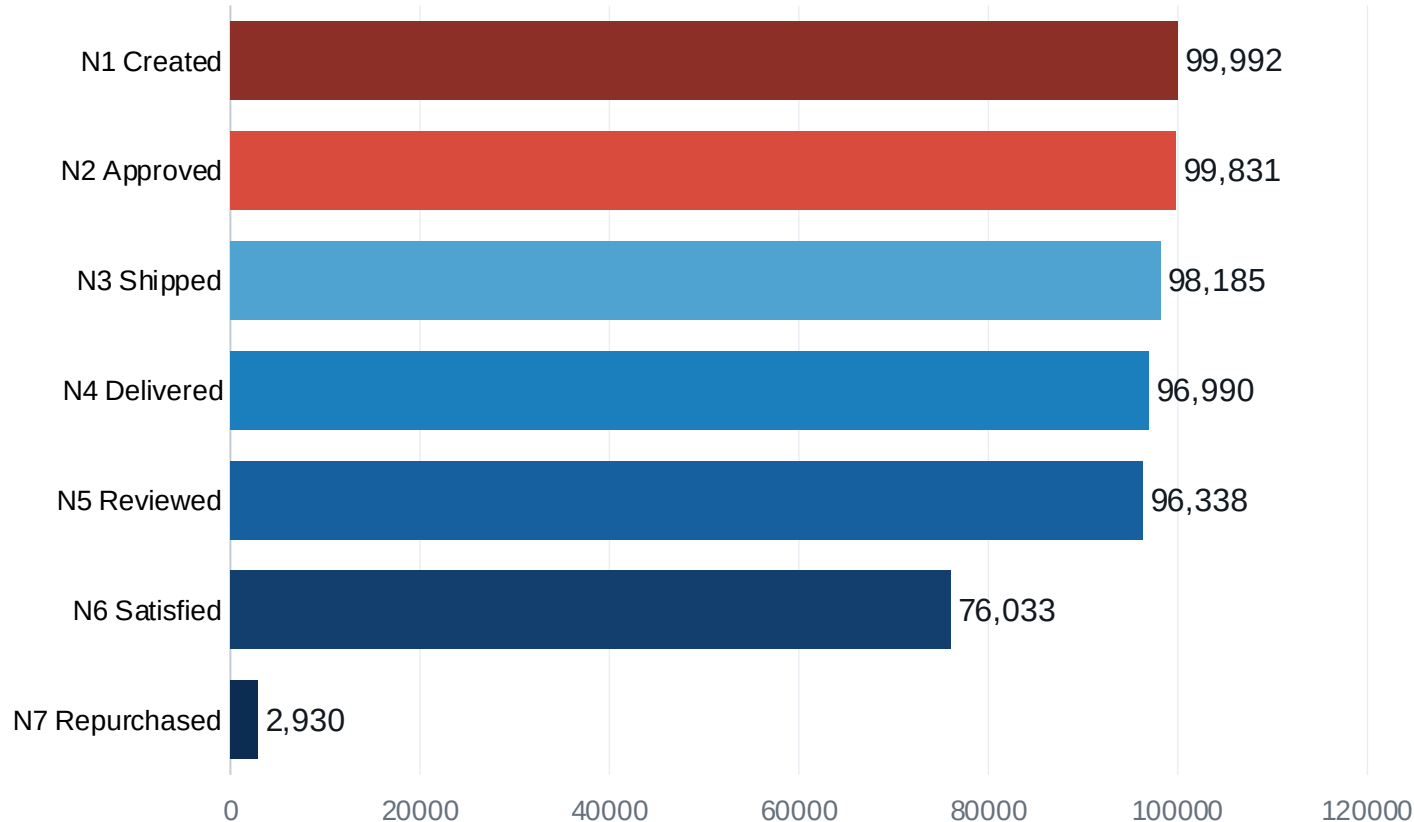
0.31% actual

Read the three tables as one sentence: Olist beats the industry at approving, shipping, delivering, collecting reviews and even satisfying buyers. It loses the game entirely at the step after satisfaction, and that single step drags the retention windows and the annual churn rate with it.

A benchmark that only says we are behind is useless. The useful reading is that everything upstream is already competitive, so the whole budget belongs at the satisfied-to-repeat step.

Upstream conversion is 99%; satisfied-to-repeat is 3.85%

Order funnel, 7 stages, 99,992 orders



Two breaks, one bottleneck

- 1 Break 1 · review → satisfaction (78.9%).** 20,305 reviewed orders score below 4 — a quality gap that upstream logistics and product QC create.
- 2 Break 2 · satisfied → repeat (3.85%).** 96% of satisfied customers never return, which is what caps revenue.

Stop optimising traffic and fulfilment: the money is in the satisfied-to-repeat step.

The last two steps lose 93,408 orders — one cliff

Absolute leakage per node over 99,992 orders, plus the worst five states and categories

Where the volume is actually lost

Leak	Start	Lost	Leak rate	% of all orders
1 → 2 Created → approved	99,992	161	0.16%	0.16%
2 → 3 Approved → shipped	99,831	1,646	1.65%	1.65%
3 → 4 Shipped → delivered	98,185	1,195	1.22%	1.20%
4 → 5 Delivered → reviewed	96,990	652	0.67%	0.65%
5 → 6 Reviewed → satisfied	96,338	20,305	21.08%	20.31%
6 → 7 Satisfied → repurchased	76,033	73,103	96.15%	73.11%

Node 5 to 6 loses 20,305 orders (21.1%) and node 6 to 7 loses 73,103 customers (96.2%). The four upstream nodes together lose under 4%. So the funnel is not leaky - it has one cliff at the end.

Worst five states by repurchase conversion

Group	Orders at N1	Reviewed → satisfi ed	Satisfied → repurch ase
SC	3,651	78.0%	3.05%
PR	5,064	81.5%	3.47%
DF	2,160	78.8%	3.65%
BA	3,397	72.8%	3.68%
MG	11,706	79.9%	3.85%

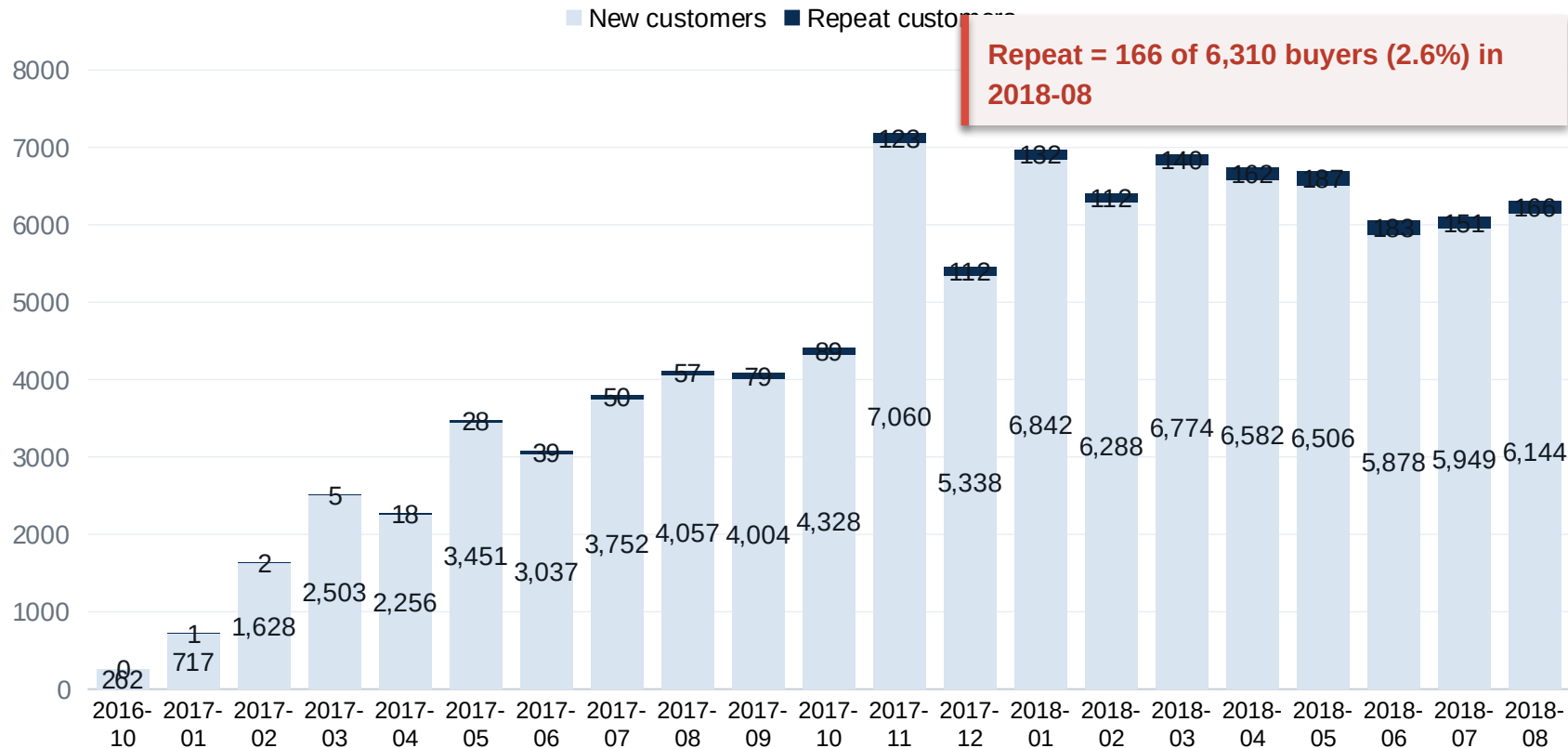
Worst five categories

Group	Orders at N1	Reviewed → satisfi ed	Satisfied → repurch ase
relogios_presentes	5,615	77.7%	2.77%
automotivo	3,900	79.6%	2.82%
telefonias	4,193	75.9%	2.95%
brinquedos	3,873	81.1%	3.24%
utilidades_domesticas	5,843	80.1%	3.46%

Percentage conversion hides the size of the prize. Counted in orders and customers, the last two nodes are the entire opportunity; fixing a 0.16% leak at node 1 is not the same job as fixing a 96% leak at node 6.

Repeat buying adds ~166 a month; new buyers add 6,000+

Buyer composition by month, 2016-10 → 2018-08



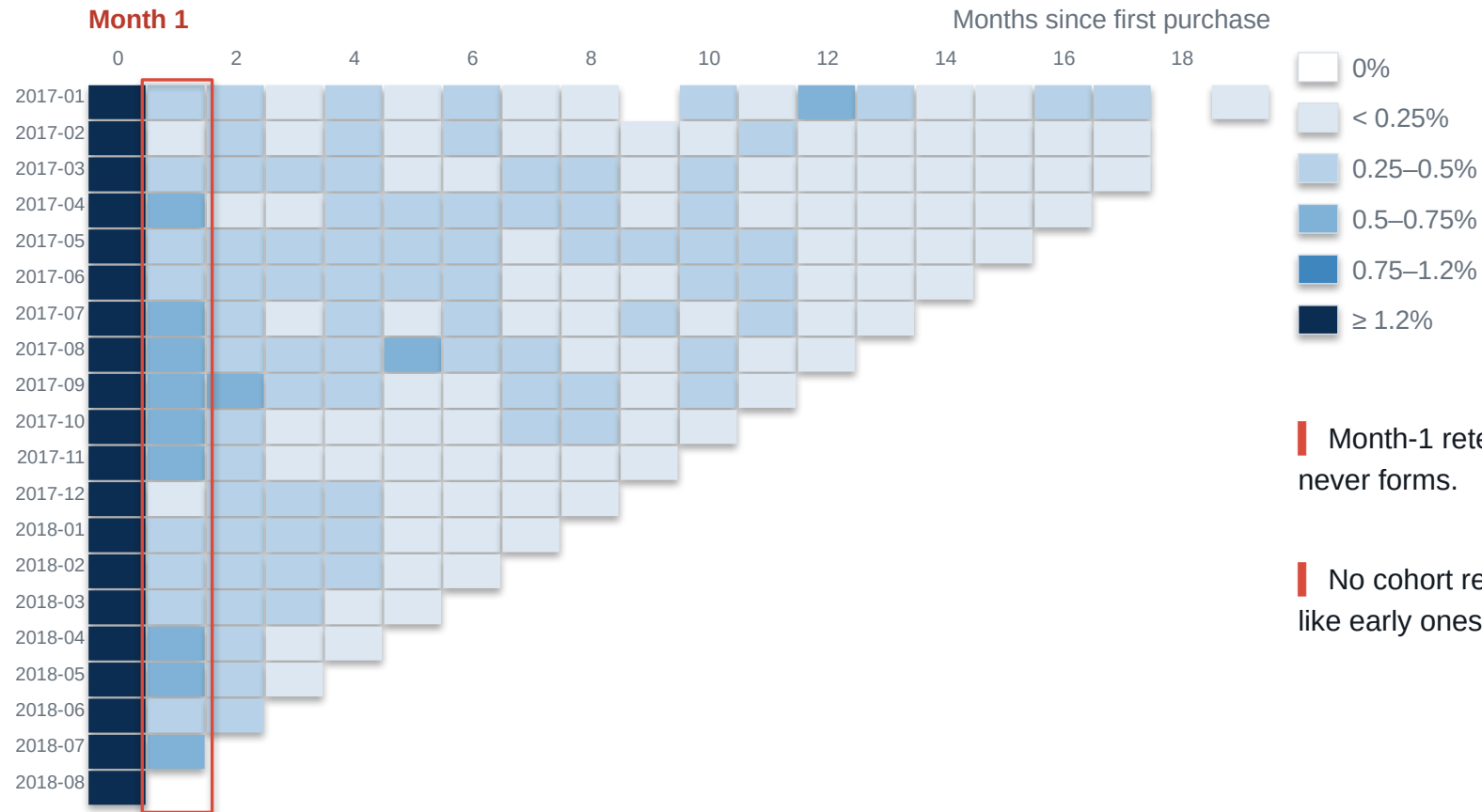
What the trend says

- Repeat buyers rise from ~0 to 166 per month — real but far too slow to compound.
- 6,000+ new buyers arrive every 2018 month, so the repeat sliver never becomes the base.
- Repeat rate stays inside a 0–3% band for 21 straight months.
- Caliber note: 29% of these repeat buyers placed every order on one day — a split cart, not a returning customer. The true cross-period repeat rate is 2.10% (S22).

This is a structural retention problem, not a bad quarter; without intervention the mix never flips.

Cohort retention falls to 0.5% in month one

Retention by first-purchase cohort × months since first purchase



Month-1 retention averages 0.5% across 20 cohorts — the habit never forms.

No cohort recovers after month 3; late cohorts behave exactly like early ones.

Cohort decay is flat, so the fix must change the first 30 days of experience rather than run seasonal campaigns.

Three of four customer classes sit outside the healthy band

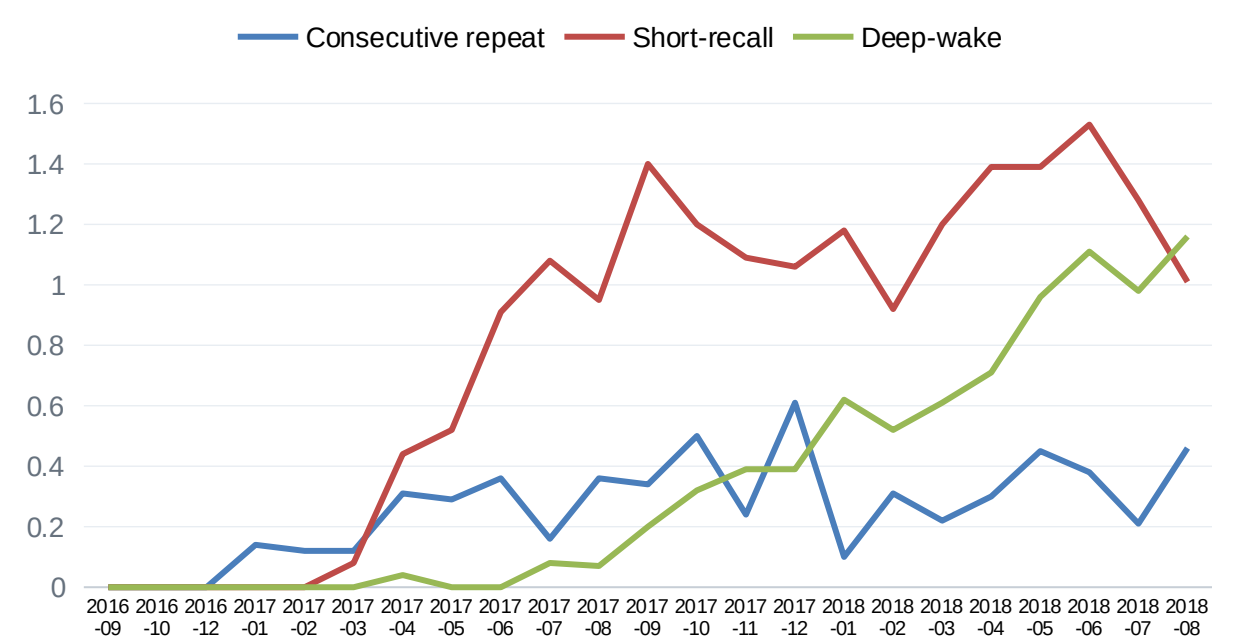
23 months, 95,194 buyer-months, benchmark ranges from cohort-analysis practice

Four classes against the healthy band

Class	Monthly avg	Healthy band	Deviation	Last month	Status
New buyers	98.07%	30–40%	+180.2%	97.37%	far outside
Consecutive repeat	0.31%	30–100%	-99.5%	0.46%	far outside
Short-recall	1.08%	15–25%	-94.6%	1.01%	far outside
Deep-wake	0.53%	0–5%	-78.8%	1.16%	healthy

New buyers never fall below 96% in any of the 23 months, so they stay out of the chart and sit in the table instead. The three returning classes together never exceed 3.9% of buyers, and the class that should carry retention value - consecutive active customers - averages 0.31% against a healthy 30-100% band.

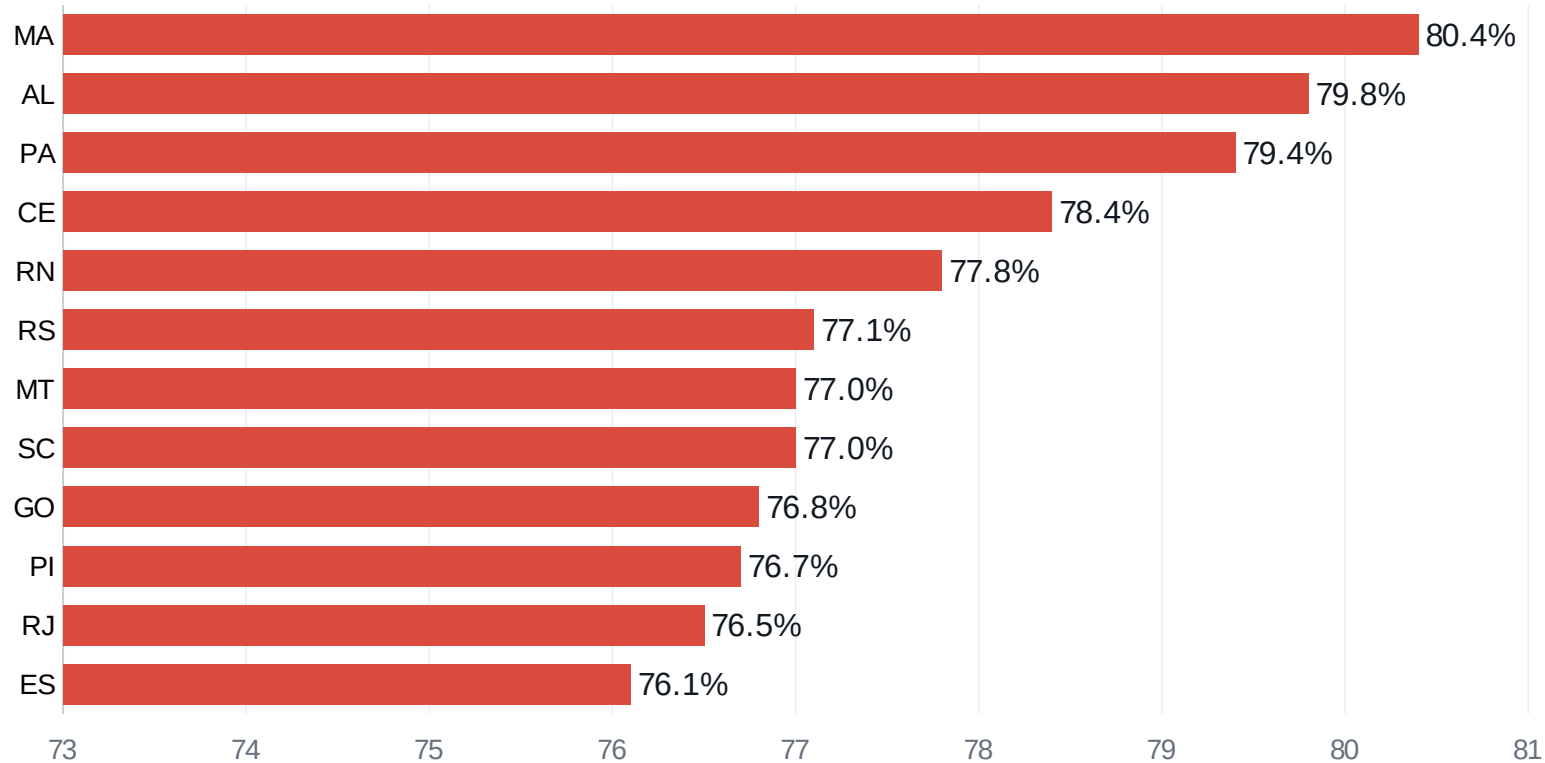
Monthly shares of the three returning classes



This is the structural version of the churn problem: the platform is not losing loyal customers, it has never produced them. That is why the fix has to start at the first repurchase rather than at a loyalty tier.

Distance from the São Paulo hub adds 7pp of churn

Actual churn rate by state, states with $n \geq 300$



What the map means

MA, AL and PA churn 79–80% versus SP's 71.3% — roughly 7pp spread.

These are the states farthest from the São Paulo fulfilment hub, so the gap is a delivery signal, not a regional preference.

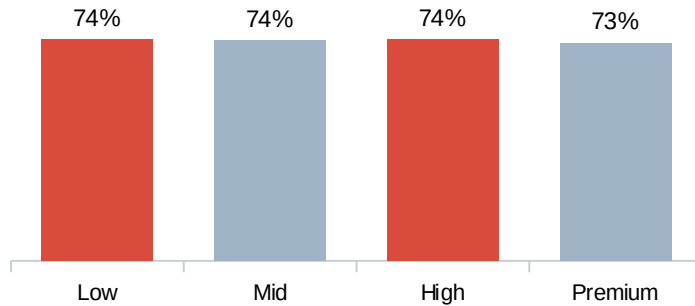
SP alone holds 39,139 customers (42% of the base), so the average is anchored there.

Route improvements should be sequenced by distance from the hub, not by customer value.

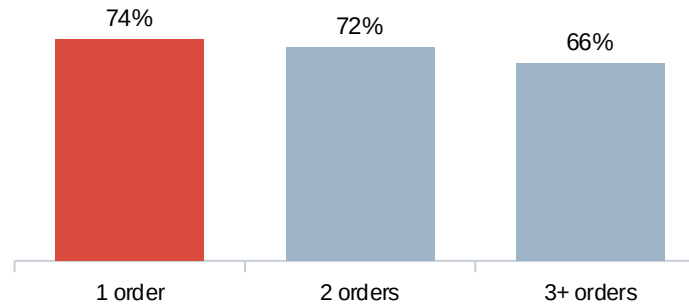
Late delivery spans a 20pp churn gap; value tier spans 1pp

Actual churn rate by delivery-experience tier (value tiers shown for contrast)

Value tier



Purchase frequency

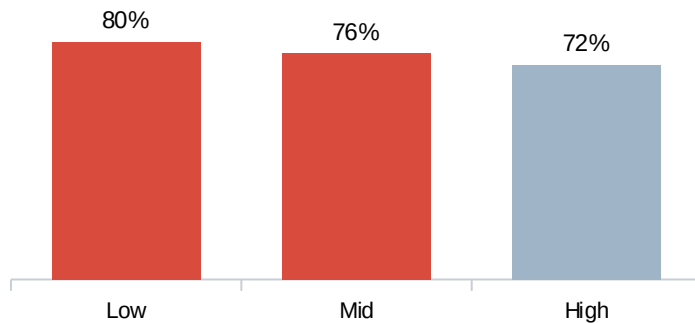


Experience moves churn; value does not

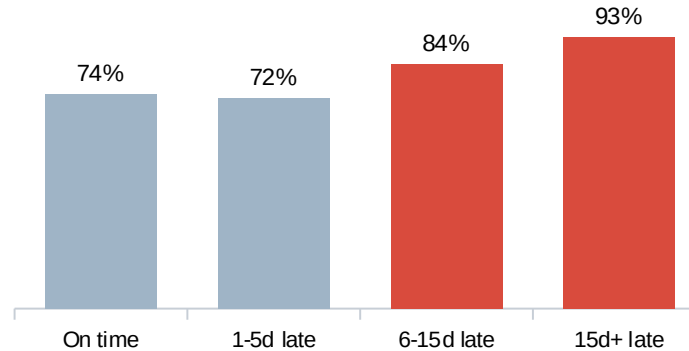
Delivery experience spans 19.7pp (73.5% on time → 93.2% at 15+ days late).

Value tier spans 1.2pp and purchase frequency only 8pp — customer value is not the churn lever.

Satisfaction



Delivery experience



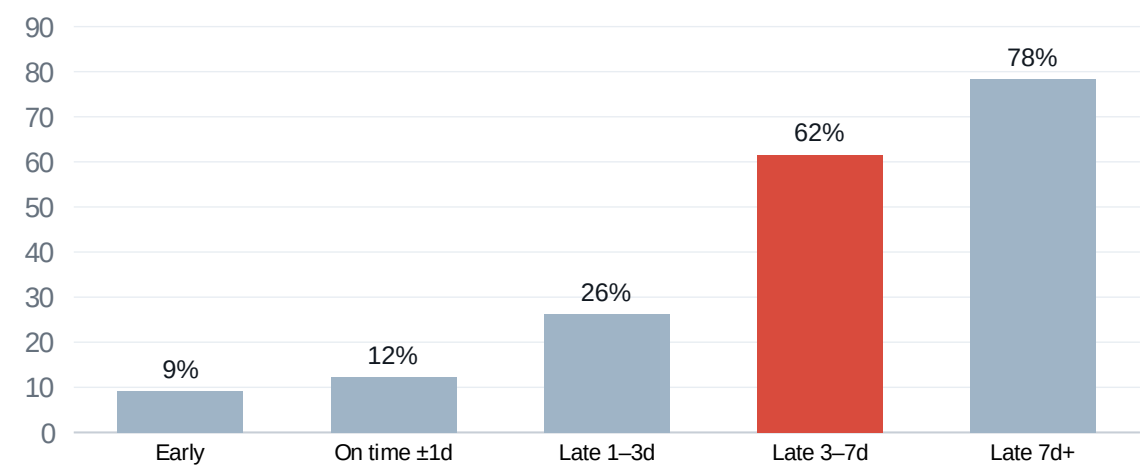
Satisfaction follows the same pattern: 79.7% churn at ≤2.5 stars versus 72.0% at >4.

Target the delivery experience of every customer rather than segmenting by spend.

Satisfaction breaks at 3-7 days late: low scores 26% to 62%

Five delay buckets over 99,992 orders, plus delivery time by category and state

Low-score share by delay bucket



Slowest five categories

Group	Avg days	Avg score	Low-score %
office furniture	20.8	3.52	25.4%
fashion shoes	15.5	4.29	10.9%
living room	13.8	3.94	17.2%
appliances 2	13.8	4.16	12.1%
consoles & games	13.6	4.14	12.1%

Delay buckets in detail

Delay bucket	Orders	Share	Avg score	Low-score %
Early	88,552	91.90%	4.294	9.2%
On time ±1d	1,431	1.49%	4.035	12.3%
Late 1-3d	1,335	1.39%	3.493	26.2%
Late 3-7d	1,790	1.86%	2.305	61.6%
Late 7d+	3,245	3.37%	1.731	78.3%

Slowest five states

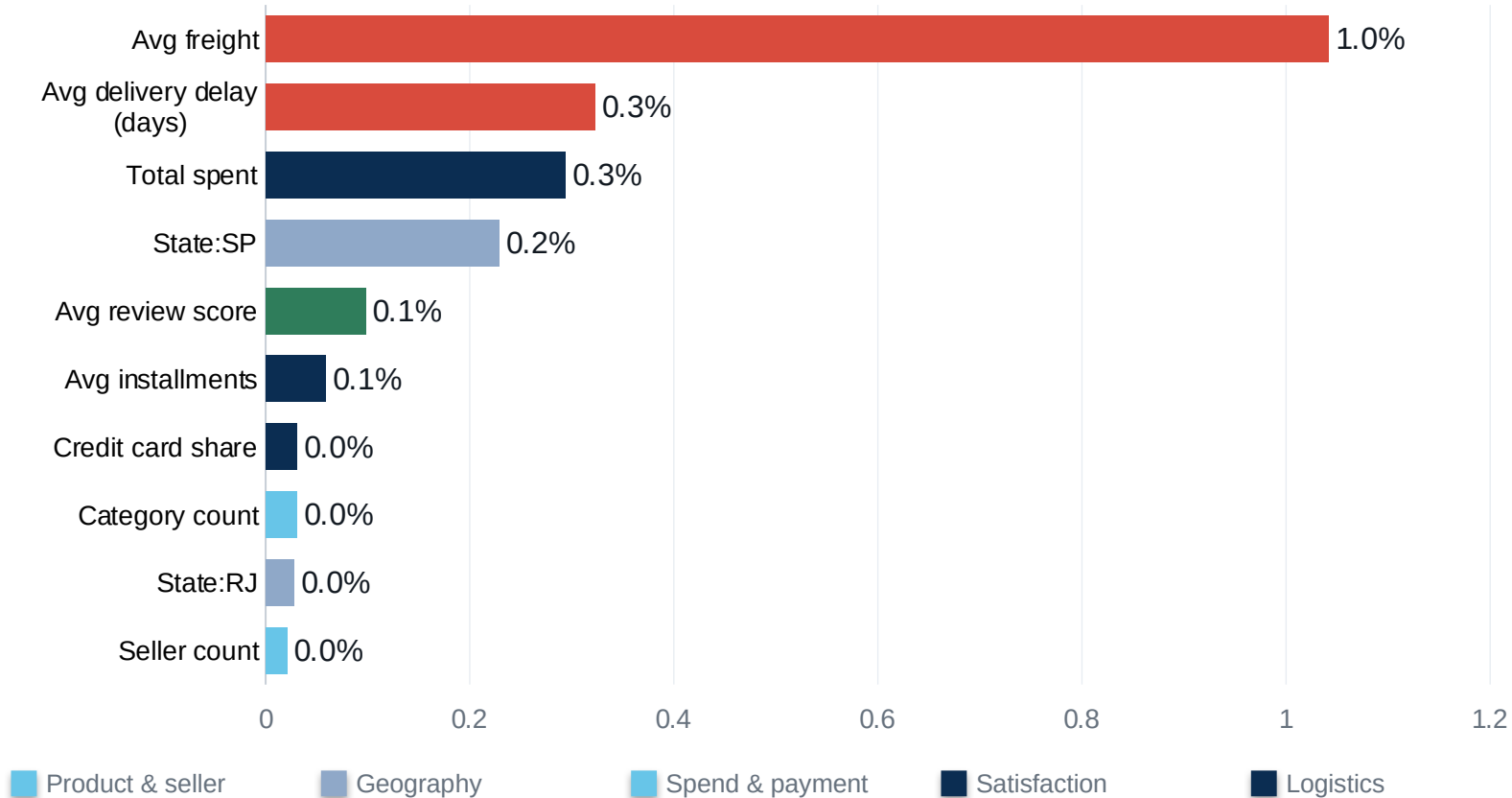
Group	Avg days	Avg score	Low-score %
AM	26.3	4.22	13.8%
AL	24.4	3.84	21.4%
PA	23.6	3.91	18.0%
SE	21.4	3.91	18.9%
MA	21.4	3.84	19.8%

Caliber warning carried on the page: on-time holds only 155 orders against 88,552 early ones, so the on-time versus early gap is not an operational target - the target is getting orders into the plus or minus one day band.

Satisfaction does not decay linearly with delay; it falls off a cliff between three and seven days. An SLA written around that cliff is cheaper than a general promise to deliver faster.

Freight and delay carry 59% of the churn signal

Global SHAP importance, top 10 of 38 features



What the model weights

Freight alone carries 45.0% of attribution (mean |SHAP| 1.042); delivery delay 14.0%.

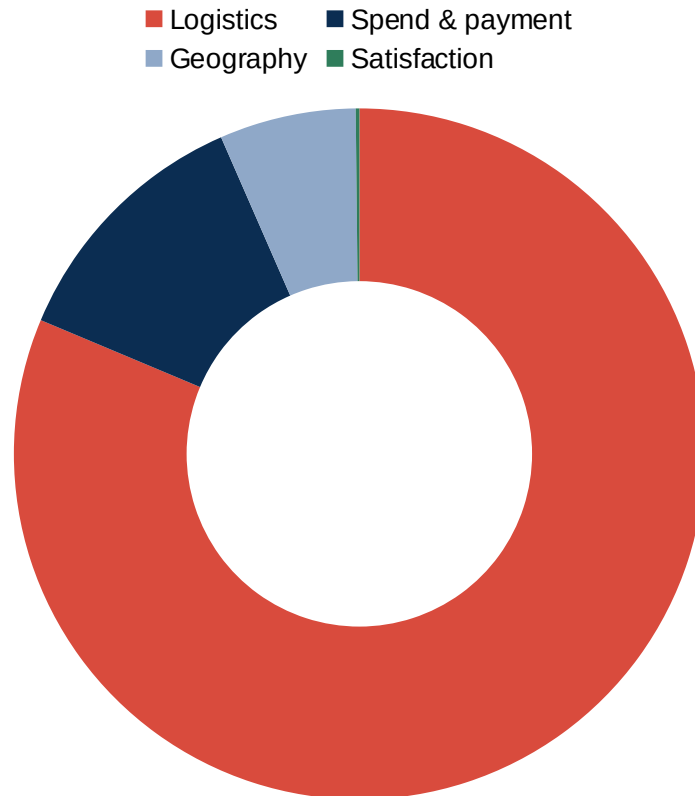
Spend (12.7%), São Paulo residency (9.9%) and satisfaction (4.2%) follow far behind.

Logistics factors total ≈59% — the single largest controllable block.

The model identifies a freight and delivery problem, not a loyalty-programme problem.

81% of customers churn for a logistics reason we control

Each customer's #1 churn driver, rolled into four categories



Rolled up per customer

Logistics	75,924	81.3%
Spend & payment	11,316	12.1%
Geography	5,960	6.4%
Satisfaction	158	0.2%

Interventions can therefore be aimed at individuals: each customer carries a named #1 driver with an impact value and direction, which is what the watchlist exposes.

Move from segment campaigns to a logistics-first individual treatment list.

Churned customers pay 22% more freight and wait 7.8 days

Churned vs retained customers, per-customer averages

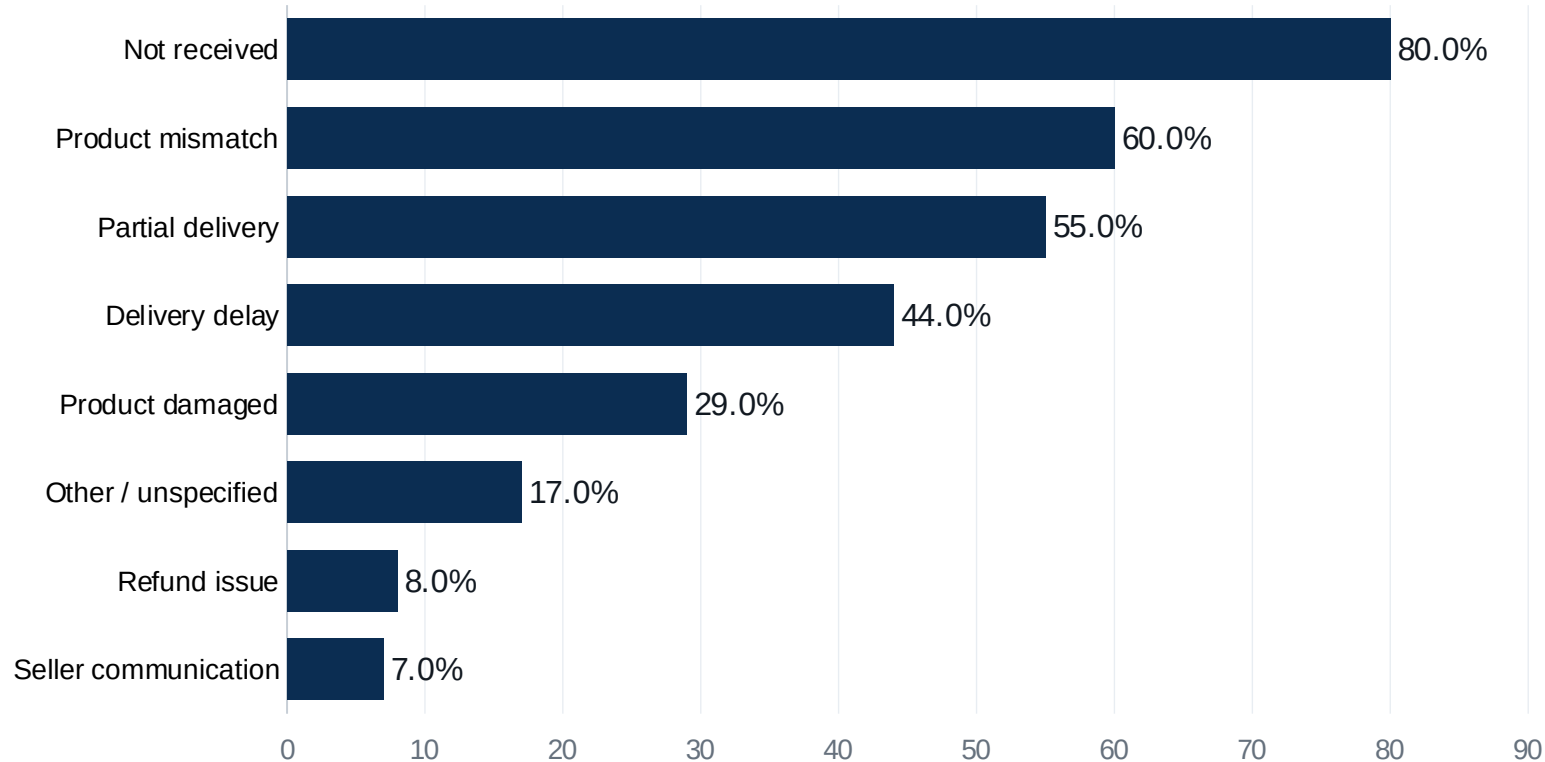
Feature	Churned	Retained	Gap
Freight per order	R\$19.69	R\$21.59	+9%
Delivery delay	-10.9 d	-11.9 d	+1.1 d
Review score	4.10	4.28	-0.18
Late delivery rate	8.8%	6.3%	+39%
Orders per customer	1.0	1.0	—

Read the table left to right: every operational difference points the same way. Churned customers pay more freight, wait longer and rate worse — and the operations team sees the same pattern in late-order scores (2.55 vs 4.22). One more check on the right-hand column: the 24,196 “retained” customers are not a loyal base — 96.7% of them have still bought only once and their basket (R\$144.96) is level with the churned group (R\$141.24). They are churners who have not expired yet, which is why retention starts at the first repurchase rather than at loyalty tiers.

The model's top drivers are levers operations already owns, so the recommendation is executable, not aspirational.

60% of complaints are logistics, confirming the model

Pain-point mix, 300 low-score reviews (rating ≤ 2)



What customers say

Not received (80), product mismatch (60), partial delivery (55) and delay (44) lead the 300-complaint sample.

Logistics complaints total 59.7% — independently the same ranking the model produced.

Only 4% of complaints are severe, so most are fixable service failures rather than lost trust.

The customer's own words give service teams the script for the top three failure modes.

The model ranks churners with 92.5% recall

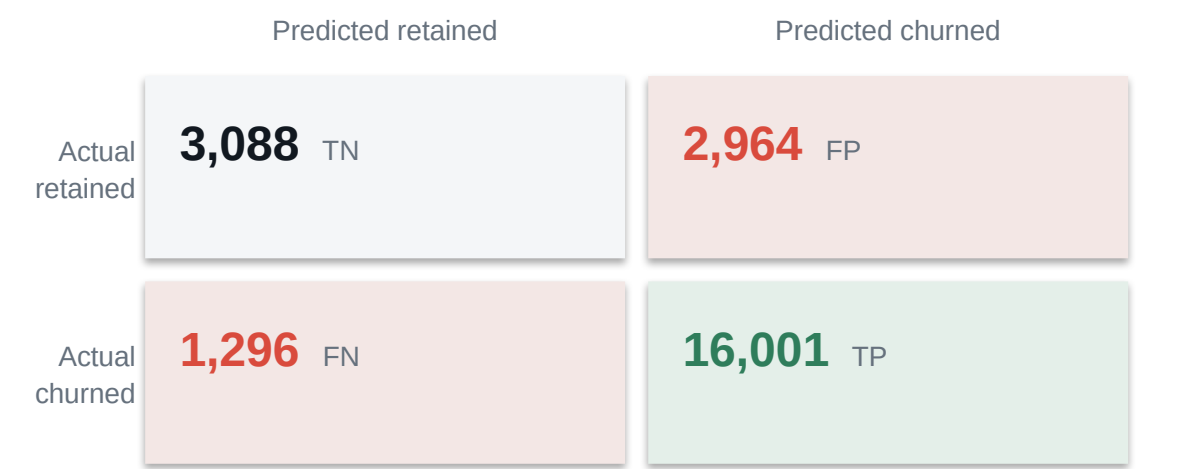
XGBoost churn model, test split n = 23,349, threshold 0.30

Test-set performance (threshold 0.30)

Metric	Value
AUC	0.868
Accuracy	0.818
Precision	0.844
Recall	0.925
F1	0.883

Recall-first threshold: a missed churner costs more than a wasted coupon, so the model flags 80.9% of customers and leaves 4 in 5 alerts actionable.

Confusion matrix (23,349 test customers)



16,001 of 17,297 churners are caught (92.5%), at the cost of 2,964 false alarms among a small retained group.

A recall-first model is the right instrument for a coupon or call campaign; it is not a churn forecast for finance.

One effect survives four estimators; the repeat link dies

70,047/23,349 split, 38 leakage-free features, PSM / DML / DoWhy / IV side by side

Model card: what the score is and is not

Item	Value
Target	no purchase for >180 days (74.08%)
Split	70,047 / 23,349
Features	38 leakage-free
Class weight	scale_pos_weight = 0.35
Threshold	0.30, chosen for F1
AUC / F1	0.868 / 0.882
Precision / recall	0.844 / 0.925
Leakage control	recency dropped (it defines the label)

0.868

AUC

test split 23,349

0.925

Recall @ 0.30

recall-first on purpose

Same question, four estimator families

Method	Question	Estimate	Verdict
PSM ATT	Late → churn	+0.0389	significant
PSM ATT	Late → satisfaction	-0.4635	significant
PSM ATT	Late → repurchase	+0.0011	not significant
DoWhy linear regression	Late → repurchase	-0.0034	≈0
DoWhy doubly robust (DML)	Late → repurchase	-0.0006	≈0
DoWhy propensity matching	Late → repurchase	+0.0000	≈0
IV 2SLS	Late → repurchase	-0.0006	not significant (p=0.61)
Logit marginal	Late → churn	+0.0299	borderline (p=0.054, OR=1.030)

38

Features

recency excluded

0.35

Class weight

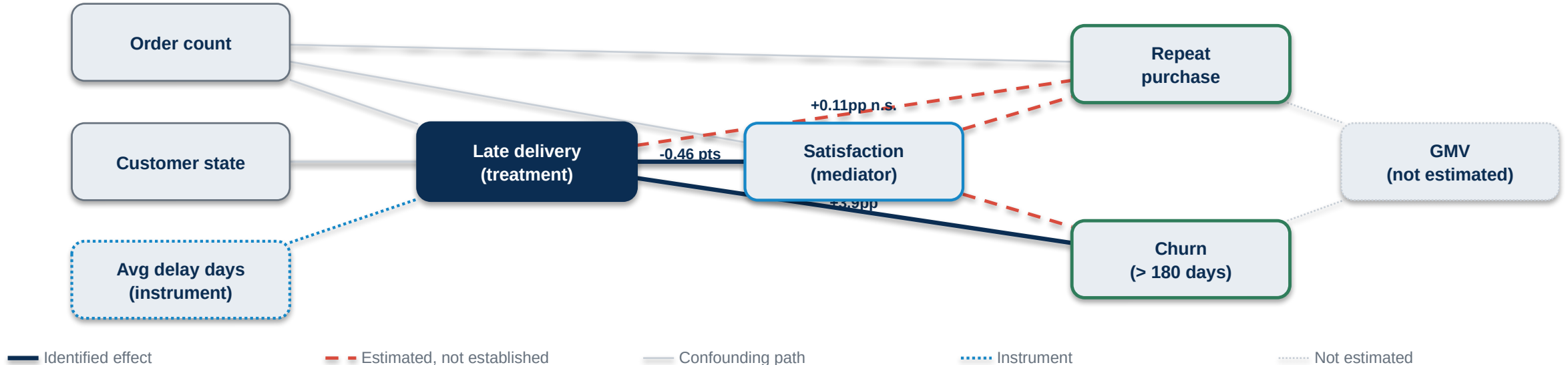
no rebalancing beyond this

Two things are being certified here. First, the score is honest: the label is defined by recency, recency is excluded from the features, and the threshold is chosen for recall because a missed churning costs more than a wasted coupon. Second, the causal claim is not a single model's opinion: churn + and satisfaction - replicate across PSM, DML, DoWhy and IV, and the repurchase link collapses to zero in all four.

The report is therefore split into what is identified (logistics to churn and to satisfaction) and what is not (anything about repurchase), and the second half is handed to a 90-day control group instead of being asserted.

Identification: three backdoor paths, two variables

DAG with the minimum adjustment set, and the refutation checks behind it



Three backdoor paths — all blocked by the adjustment set

Late delivery $\leftarrow$ Order count $\rightarrow$ Satisfaction
 Late delivery $\leftarrow$ Customer state $\rightarrow$ Satisfaction
 Late delivery $\leftarrow$ Order count $\rightarrow$ Repeat

Minimum adjustment set

{OrderCount, CustomerState} $\rightarrow$ all three paths d-separated

Estimand selection 3 backdoor paths, 2 adjustment variables · avg delay days · $p = 0.61$, not significant · mediator unobserved in part

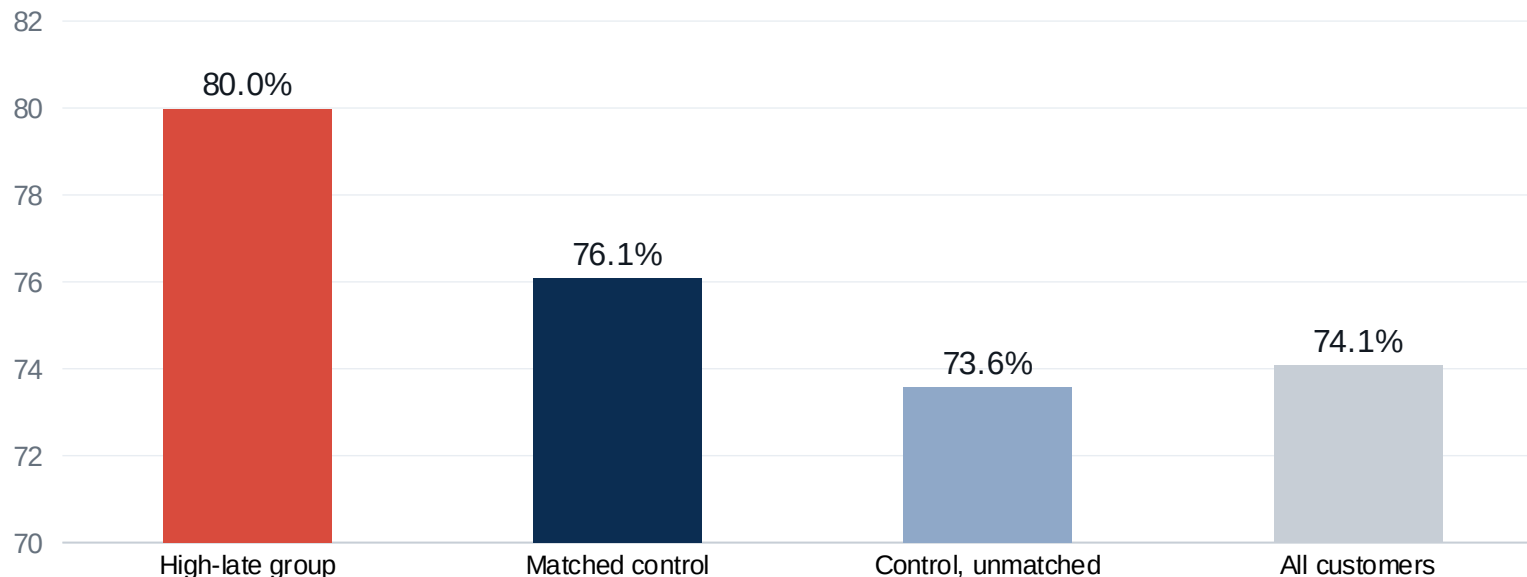
Robustness

Placebo / 50% subsample / random-cause refuters all pass (3/3; placebo ATE $4.1e-05$ vs -0.0034 original)

The identification is an argument, not an assumption: name the paths, block them with two variables, then prove the null with refuters.

Late delivery causes the churn: +3.9pp, not the raw 6.4pp

Raw vs identified effect, customers matched on order count, spend and state (n = 93,396)



What is identified, and what is not

- Treatment is customer-level: 8.0% of customers (7,497) had a majority of orders delivered late.
- Satisfaction moves the same way but shrinks harder: -1.72 raw points become -0.46 once matched (27% retained).
- The repeat link does not survive: IV $p = 0.61$, DML -0.06pp, and the matched sign flips. That is why programme ROI is a band, not a number.
- Robustness: placebo, 50% subsample and random-cause refuters all pass (placebo ATE $4.1e-05$ versus -0.0034 original).

How much of the raw gap is real

· value axis starts at 70%; bars are levels, the effect is the difference

Raw difference

+6.4pp

High-late vs all other customers

Removed by matching

-2.5pp

Different customers, not different delivery

Identified effect

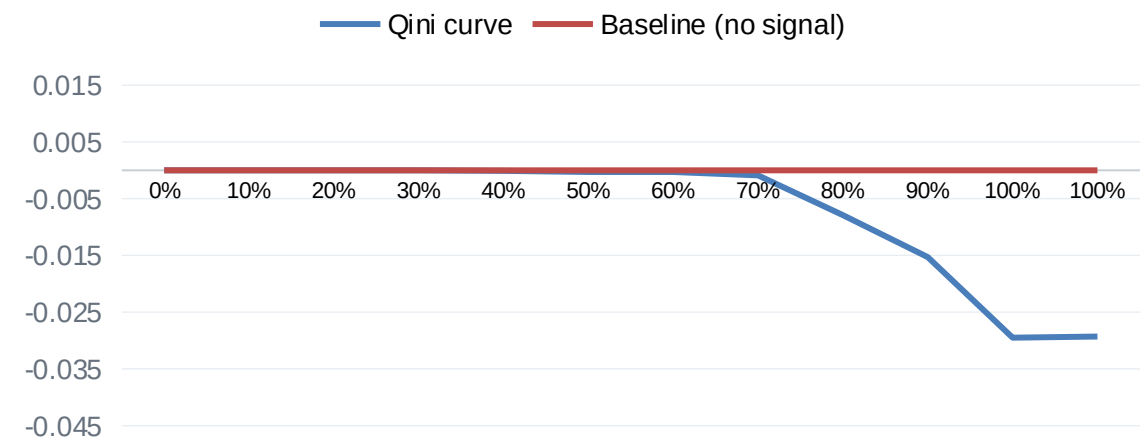
+3.9pp

61% of the raw gap survives

Fund the input we control — the delivery experience — and settle the repeat link with the 90-day control group rather than with an assumption.

Heterogeneity is real; the ranking is not (Qini -0.03)

Qini / AUUC and the decile check on the T-Learner holdout (n = 8,997)



Deciles ranked by predicted uplift

Decile	Pred uplift	Treated n	Events	Actual repeat
1	-0.065	787	6	0.76%
2	-0.011	834	5	0.60%
3	+0.061	601	1	0.17%
4	+0.119	14	0	0.00%
5	+0.132	1	0	0.00%
6	+0.146	2	0	0.00%
7	+0.162	3	0	0.00%
8	+0.186	3	0	0.00%
9	+0.220	1	0	0.00%
10	+0.255	3	0	0.00%

-0.029
Qini curve, end
baseline 0

-0.0035
AUUC
target > 0.05

12 : 234
repeat events, treated : control
20:1 imbalance

7 / 0
treated customers / events in D4-D10
no validation base

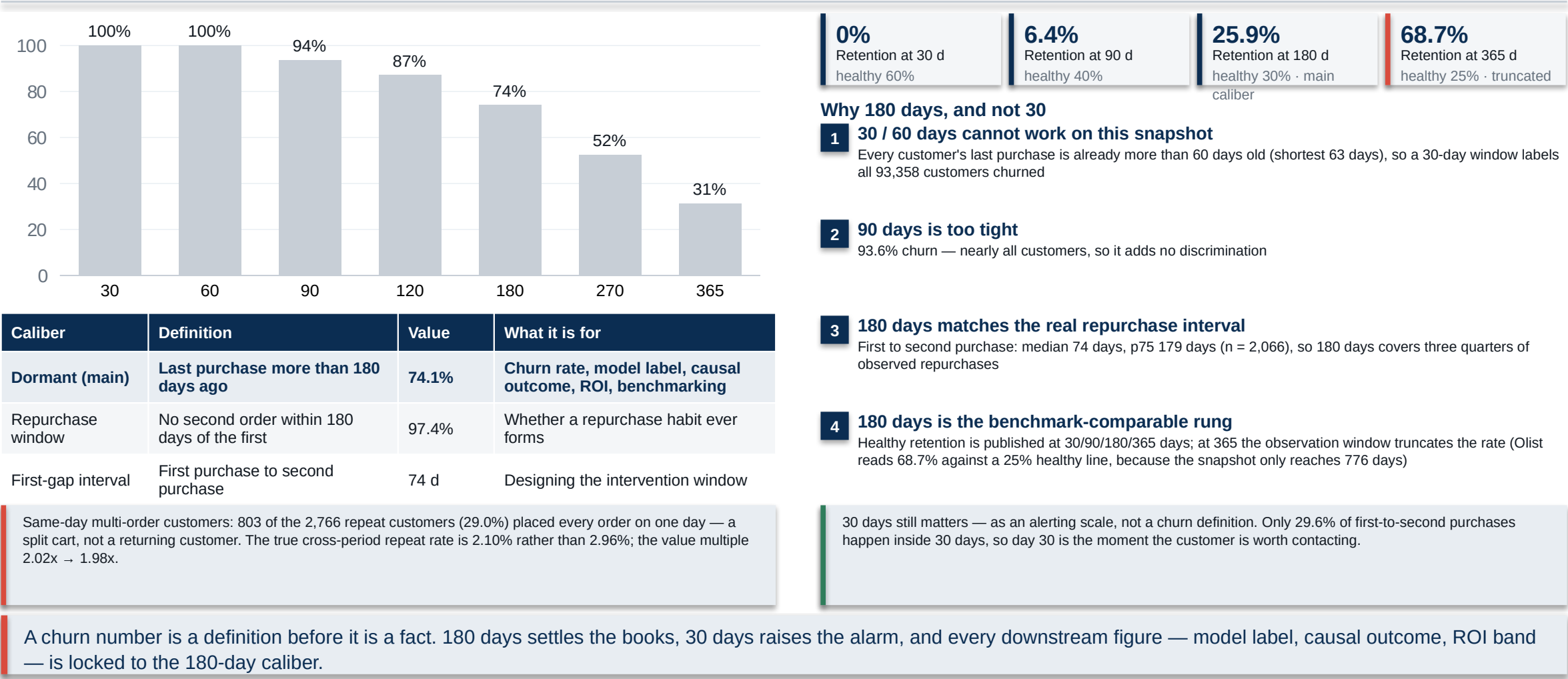
Why it fails: sample, not direction
The outcome is extremely sparse: repeat runs at 2.96% overall and only 12 of 2,249 treated customers repeat — a 20:1 event imbalance against 234 control events. The T-Learner's CATE is therefore unstable and Qini/AUUC fall below baseline: a precision problem, not an absent effect (PSM, DML and IV agree). Caveat: `class_weight='balanced_subsample'` on 12 positives inflates treated-arm probabilities — the source of the +31pp phantom uplift

How it gets fixed
Widen the event base: use a compound repeat/retention outcome inside a 180-day window, or extend the observation window · Re-estimate CATE on a randomised holdout (A/B) instead of observational counterfactuals · Drop `class_weight` and move to an X-Learner / DR-Learner with probability calibration · Re-accept on both the Qini/UAUUC curve and decile monotonicity before any list ships

The metric, the diagnosis and the fix ship; the targeting list does not — a ranking that fails its own Qini test costs more than no ranking.

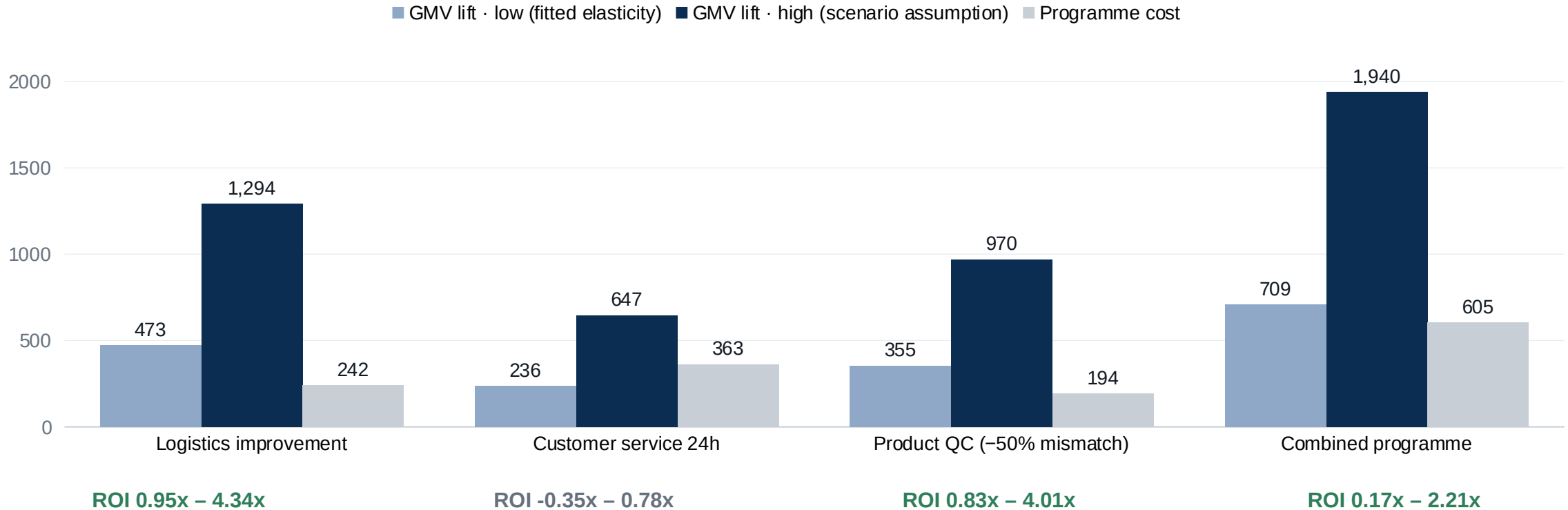
Why 180 days: 74% churn at 180, 94% at 90, 100% at 30

Caliber sensitivity across 30/90/180/365-day windows, 93,358 customers



Logistics: 0.95–4.34x; clears cost only at the top

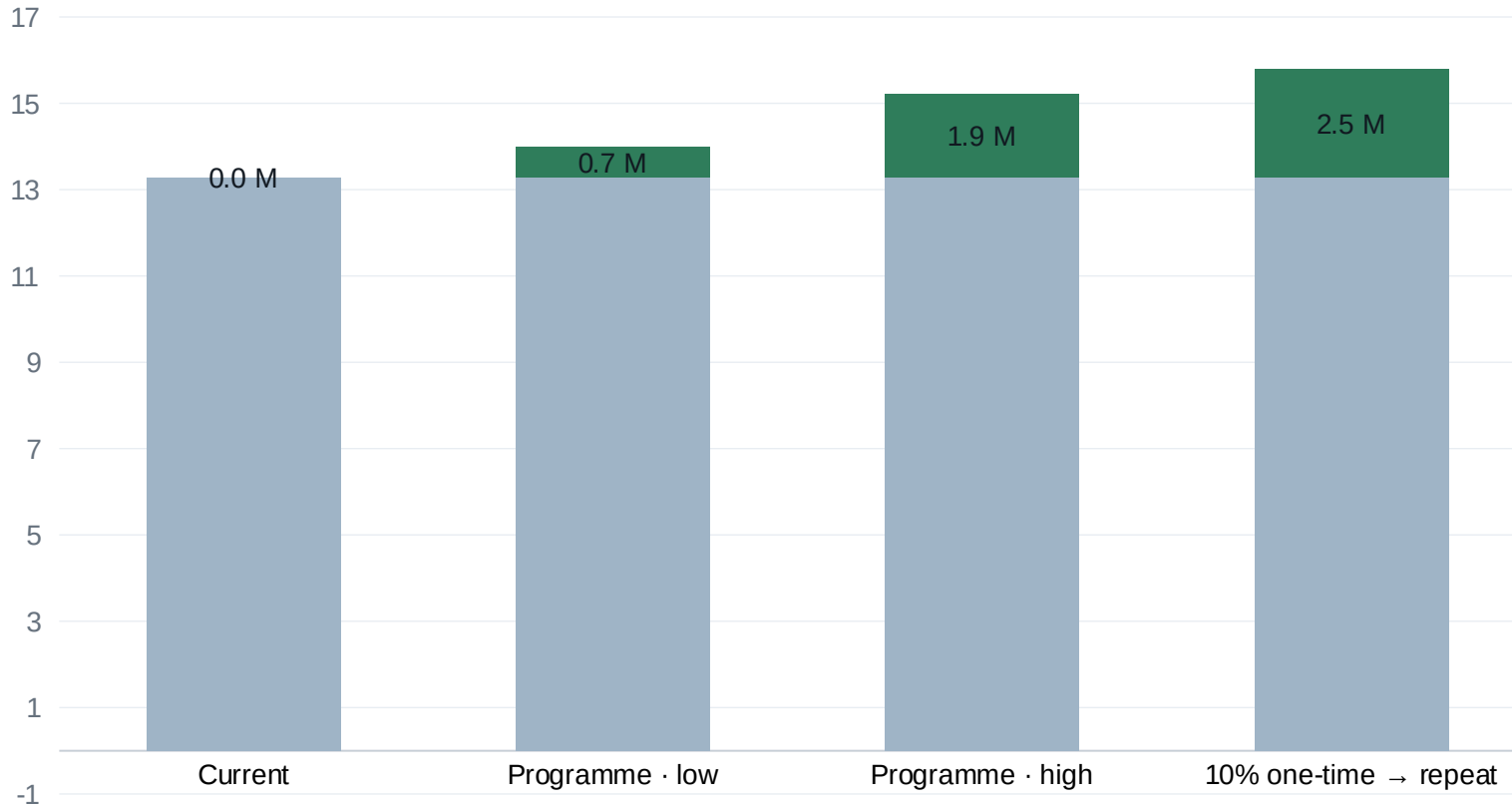
Annual GMV lift under a 1.46–4.00 pp repeat elasticity band vs programme cost



At the fitted lower bound none of the four clears its cost; logistics and product QC reach payback only at the upper bound. The control group settles which end applies.

Worth R\$0.7–1.9M a year at 0.17x–2.21x its cost

Two independent estimates of the same opportunity, annualised



Two routes, one decision

■ Programme route: R\$0.7–1.9M annual GMV for R\$605k, i.e. 0.17x–2.21x — below cost at the fitted bound, payback only at the assumption.

■ Demand route: converting 10% of one-time buyers at R\$277.92 adds R\$2.5M, i.e. +19% of revenue.

■ Retention spend is self-funding: keeping a customer costs 5–10× less than acquiring a replacement.

Both routes point at the same opportunity; the programme route is the one we can start on Monday, and the control group decides where inside the band it lands.

A 90-day sequence turns diagnosis into measured gains

Three phases, owner-level actions, measurement at every gate

Days 1–30 · Instrument

- Stand up the ten-page dashboard on the client warehouse.
- Agree the churn definition, threshold and owners.
- Baseline funnel, cohort and repeat-rate metrics.

Gate: Baseline signed off; watchlist live for CRM.

Days 31–60 · Act

- Run the logistics programme on the worst states (MA, AL, PA, RO).
- Fix the top three product-mismatch defects with sellers.
- Launch the high-risk list into service and coupon flows.

Gate: Interventions live in the four worst states.

Days 61–90 · Prove

- Hold out a control group and read measured uplift.
- Re-estimate ROI with measured, not assumed, lift.
- Automate refresh and churn-rate drift alerts.

Gate: Measured uplift and revised ROI reviewed by the board.

Ninety days is enough to prove the mechanism; the dashboard keeps it honest afterwards.

Ten Power BI pages keep churn under weekly watch

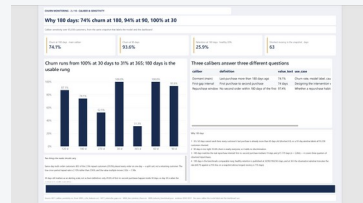
10 pages · 31 tables · 165 measures · 171 visuals · PBIP/TMDL/PBIR, refreshable

1 · Churn overview



dashboard capture

2 · Churn caliber



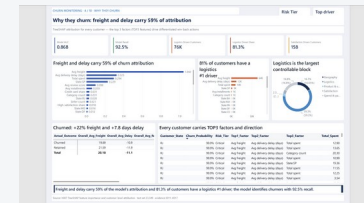
dashboard capture

3 · Who stays, who leaves



dashboard capture

4 · Why they churn



dashboard capture

5 · Cost of delay



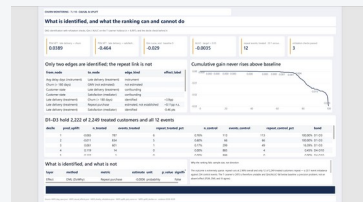
dashboard capture

6 · Review themes



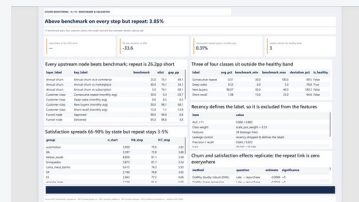
dashboard capture

7 · Causal & uplift



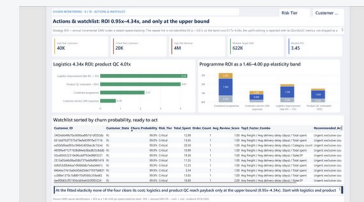
dashboard capture

8 · Benchmark · structure · validation



dashboard capture

9 · Actions & watchlist



dashboard capture

10 · Appendix · order funnel & structure



dashboard capture

The analysis becomes a control loop only if someone owns the weekly read of page 1 and page 5, and a quarterly read of page 6.

Definitions and limits: what is measured and where it stops

Churn definition, snapshot coverage, model scope, source ledger

Metric dictionary: what each number means

Metric	Definition and caliber
Churn	No purchase for more than 180 days; 74.1% of customers. Main caliber: churn rate, model label, causal outcome, ROI.
Caliber sensitivity	Same customers at 30/90/180/365 days: 100% / 93.6% / 74.1% / 31.3%. See S22.
Retained customer	Bought within 180 days: 24,196 customers, 26.4% of revenue, R\$144.96 each; 96.7% still bought only once.
Repeat rate	Buyers with an earlier purchase ÷ all buyers; 2.96%, or 2.10% excluding same-day split carts.
Cohort retention	Share of a first-purchase cohort still buying in month N; month one is 0.5%.
Model and threshold	XGBoost holdout: AUC 0.868, recall 0.925, F1 0.883; threshold 0.30 (F1-optimal).
Identified effect and uplift	Late delivery → churn +3.9pp, satisfaction -0.46 (PSM ATT). Uplift fails: Qini -0.029, AUUC -0.0035.
ROI	(annual GMV lift – programme cost) ÷ cost, across a 1.46–4.00 pp elasticity band.

Limits the reader should carry

- No gender or age fields exist in the dataset; behavioural tiers substitute for demographics.
- The analysis uses a snapshot (99,992 orders) while the live database holds 99,441; dashboard tables are snapshot-consistent.
- Review-dependent funnel nodes (reviewed / satisfied) are indicative and depend on review coverage.
- Causal claims come only from the identified models; the repeat-rate link is not identified, so programme ROI is a band.
- The churn caliber is a definition, not a fact: 30 days marks every customer churned and 365 days truncates, so 180 days leads.

Source ledger (all local, reproducible)

h003 RFM · h004 funnel · h005 monthly structure · h006 cohorts & window curve · h007 XGBoost + SHAP · h008 review NLP · h009 causal uplift · h011 caliber & repurchase cadence.
 Power BI project: 10 pages · 31 tables · 165 measures · 171 visuals (PBIP/TMDL/PBIR).
 All figures in this deck are regenerated from these artefacts; no external data was introduced.

Decision required: fund the logistics-first retention sprint

Three decisions, named owners, 90-day milestones

Decision 1

Fund the 90-day logistics-first retention sprint.

Owner

COO with marketplace operations

By

Sprint start within 2 weeks

Decision 2

Adopt the scored watchlist as the standard operating list.

Owner

Head of CRM / customer service

By

Live in 30 days

Decision 3

Approve the control group and measurement budget.

Owner

Analytics lead with finance sign-off

By

Designed before launch

The analysis is complete and reproducible; the R\$0.7–1.9M band is narrowed by deciding to measure, not by further study.

Analysis complete; a R\$0.7–1.9M band is left to execution

What ships · what is needed · how to start

What ships with this analysis

- A ten-page Power BI churn dashboard (English and Chinese editions) on the client warehouse.
- The customer-level model results table — 93,358 customers with scores and TOP3 drivers.
- This deck plus the reproduction scripts behind every number.

■ Human-led method. AI-assisted execution: problem framing, analysis design, method selection and validation are mine; AI handled implementation and drafting, and every figure is reconciled by an automated gate before it ships.

■ A control group for the 90-day measurement.

■ Sign-off on the churn definition and the 0.30 threshold.

The analysis is reproducible end to end; the R\$0.7–1.9M band and the decision to close it with a control group are the only missing inputs.

How to start

1. Kick-off within two weeks of approval.
2. Logistics and product QC first; service spend second.
3. Review measured uplift at day 90, then re-cut the ROI.

Zhao Zhang · Data analyst · 2026-09

zzhangzhao@hotmail.com

[Upwork profile](#) 

Available for retention diagnostics, Power BI delivery and causal measurement.